Assignment #8 for Numerical Analysis

Deadline: May 8

1. (40 points)

   (a) Develop a second-order formula $D_h^{(1)} f(x)$ for approximating the first derivative $f'(x)$ that uses the data $f(x-h), f(x)$, and $f(x+3h)$ only by the following two methods.

   (i) Using the interpolation polynomial $p_2(x)$ and using $p_2'(x)$ to approximate $f'(x)$.

   (ii) Using the method of undetermined coefficients.

   (b) How many **first-order** formula $D_h^{(1)} f(x)$ for approximating the first derivative $f'(x)$ that uses the data $f(x-h), f(x)$, and $f(x+3h)$ only. Write one down if possible.

2. (20 points) Find the **error term** and **order** for the approximation formula

$$f'(x) \approx \frac{4f(x+h) - 3f(x) - f(x-2h)}{6h}.$$

3. (40 points) Consider

$$\int_0^\pi x \cos x dx.$$

   (a) Approximate the integral using the composite Midpoint Rule with $m = 1, 2, 3$. Compute the error by comparing with the exact value from calculus.

   (b) Approximate the integral using the composite Trapezoid Rule with $m = 1, 2, 3$. Compute the error by comparing with the exact value from calculus.

   (c) Approximate the integral using the composite Simpson's Rule with $m = 1, 2, 3$. Compute the error by comparing with the exact value from calculus.